Literature Review for Master Thesis

1 Nomenclature

The following symbols are used throughout the text. Units are given where applicable.

t_o : Subfault onset time [s], e.g. $t_o = (x - x_{\text{start}})/v_{\text{rupture}}$.

x, x_{start} : Horizontal coordinate and rupture starting position [m].

v_{rupture} : Prescribed rupture propagation speed [m s^{-1}].

t_{rise} : Rise time of a subfault/source [s].

Δx : Subfault width or discretisation length in x [m].

x_{slip} : Prescribed slip on a subfault [m].

M_0 : Scalar seismic moment per subfault [N m].

m_{ij} : Moment tensor components [N m].

r_{mode} : Mode mix parameter [dimensionless].

l_{cz} : Cohesive process (fracture) zone length [m].

l_{ch} : Irwin characteristic length [m].

$\lambda_{\min}$: Shortest wavelength to resolve [m].

$v_{\min}, v_{\max}$: Minimum and maximum relevant wave speeds in the model (m s^{-1}).

Δt : Time step [s] for explicit time integration.

ξ : Normalized opening [dimensionless].

δ : Local opening (separation) displacement across the interface [m].

δ_c : Mixed-mode critical opening [m] used to nondimensionalize openings.

$\delta_{n,c}$, $\delta_{s,c}$ Single-mode critical openings for Mode I (normal) and Mode II (shear) [m].

$T(\xi)$: Cohesive traction as a function of normalised opening [Pa or N m^{-2}].

T_{peak} : Peak cohesive traction [Pa].

$\phi(\Delta)$ Mixed-mode cohesive potential [J m^{-2}] with $\Delta = (\Delta_n, \Delta_t)$.

Δ_n, Δ_t : Normal and tangential separations [m].

δ_n, δ_t : Normal and tangential length scales in the cohesive potential [m].

q Coupling parameter in the mixed-mode cohesive potential [dimensionless].

ϕ_n, ϕ_t : Work (energies) of separation for pure normal and tangential modes [J m^{-2}].

$\sigma_{\max}$, $\tau_{\max}$: Peak normal strength and peak shear strength [Pa].

σ_n, τ_s Normal and shear strength [Pa].

G_I, G_{II} : Fracture energies for Mode I and Mode II [J m^{-2}].

σ_n, τ_s : Nominal normal and shear strengths [Pa].

v_s, v_p : Pressure and Shear seismic wave speeds [m s^{-1}].

ρ : Mass density [kg m^{-3}].

μ, λ Lamé parameters [Pa]

E : Young's modulus [Pa]

l_{crack}, x_0 : Local crack process length and initial crack front position [m] used in local coordinate definitions.

2 Introduction for Thesis

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that avalanches are a significhant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain that an avalanche is possible. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

In this master's thesis, the aim is to detect avalanche slab breakoff and crack propagation using modeled data from distributed acoustic sensing (DAS) cables.

In this thesis, a fibre-optic cable installed on the slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (Parker et al., 2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (Shatalin et al., 1998). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (Hartog, 2017). The phase variations are then converted into dynamic strain and from strain, into deformation along the cable (Trabattoni et al., 2023). mention also the cable location here, length etc

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (Paitz et al., 2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied.

A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Bartelt & Lehning, 2002; Brun et al., 1992). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Bartelt & Lehning, 2002; Brun et al., 1992). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the distinct element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall & Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall & Strack, 1979). This method has been used by Bobillier et al., 2020, 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall & Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (Knight & Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl & Weißgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl & Weißgraeber, 2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (Rosendahl

& Weißgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and Weißgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (Bergfeld et al., 2025; Bobillier et al., 2024; Trottet et al., 2022). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (Trottet et al., 2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (Trottet et al., 2022).These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (Bergfeld et al., 2025; Trottet et al., 2022). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2015, 2018; Gaume et al., 2017). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak & Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak & Gaume, 2019). This is in contrast to rapid loading, which happens for example due to a skier (Mulak & Gaume, 2019).

In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasiev et al., 2017).

The different snow processes described have differenit implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seimsic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

here i elaborate the research questions, but I don't know which one I will focus on yet

- Different DAS signatures for different propagation modes (sub-raleigh vs supershear)
- models based on loading rate to see if DAS data looks different
- Distinction of processes: Weak layer colapse, crown and staunchwall fractures
- Relation to snow properties in models
- relationship between crack propagation and release area

3 Methods - Cohesive Zone Model

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following Mahajan and Joshi, 2008. This model was chosen because there is a narrow band ahead of the fracture zone, the cohesive zone, where the material is already weakened. This means that before a point in the material cracks, there is already weakening of the material before. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer. Material parameters for the snow layer are taken from Guillemot et al., 2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

The mesh is generated to resolve frequencies up to the 95th percentile of the source wavelet's frequency content. Cohesive tractions are prescribed on zero-thickness interface elements and depend on the normalized opening $\xi = \frac{\delta}{\delta_c}$. The zero-thickness interface elements are implemented by not solving for volumetric strain but displacement between interfaces instead. For the linear softening law used here the traction in a given cohesive element follows

$$T(\xi) = T_{\text{peak}}\,(1-\xi) \quad \text{for } 0 \le \xi \le 1, \qquad T(\xi) = 0 \text{ for } \xi > 1,$$

where T_{peak} is the peak cohesive traction and δ_c the critical opening. The normalized opening can be related to the local crack coordinate as $\xi = \frac{(x-x_0)}{l_{crack}}$ for a crack front with initial position x_0 and local process length l_{crack}.

The mixed-mode potential used in the code follows the form from Mahajan and Joshi, 2008. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the potential is written as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right)\left\{\left[1+\frac{\Delta_n}{\delta_n}\right] + q\left[1-\exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right]\right\},$$

from which the normal and tangential tractions are obtained, the implementation uses the same closed-form expressions. This is so that peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear works of separation relate to peak strengths via

$$\phi_n = e\,\sigma_{\max}\,\delta_n, \qquad \phi_t = \sqrt{\tfrac{e}{2}}\,\tau_{\max}\,\delta_t,$$

and the critical openings for pure modes are computed as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

with the mixed-mode critical opening

$$\delta_c = \sqrt{(1-r_{\text{mode}})\,\delta_{n,c}^2 + r_{\text{mode}}\,\delta_{s,c}^2}\,.$$

Lamé parameters are derived from input seismic speeds and density: $v_s = \sqrt{\frac{\mu}{\rho}}$ so that $\mu = \rho v_s^2$, and $v_p = \sqrt{\frac{(\lambda+2\mu)}{\rho}}$ so that $\lambda = \rho(v_p^2 - 2v_s^2)$. Young's modulus is then computed from the Lamé parameters: $E = \dfrac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$ (Aki & Richards, 1981). These conversions are applied for every material in the model.

The critical opening distances for pure normal (Mode I) and shear (Mode II) separations are computed from the corresponding fracture energies, strenghts, and peak tractions as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

and combined for mixed-mode behaviour using

$$\delta_c = \sqrt{(1 - r_{\text{mode}})\, \delta_{n,c}^2 + r_{\text{mode}}\, \delta_{s,c}^2},$$

where $r_{\text{mode}} \in [0, 1]$ controls the mode I/II mix, where 0 is pure mode I and 1 is pure mode II. In the current implementation the slip on each subfault follows the prescribed linear slip law up to the critical slip distance. The mode mix is an user-input parameter, while all the other parameters are derived from laboratory experiments from Mahajan and Joshi, 2008.

The characteristic Irwin length for shear is calculated as $l_{ch} = \frac{E_{\text{eff}} G_{II}}{\tau_s^2}$, and the cohesive zone length is taken as $l_{cz} = \frac{\pi}{8} l_{ch}$, which sets the spatial extent over which cohesive weakening is active ahead of the crack tip.

Rupture propagation speed is chosen with respect to the local shear-wave speed and the bounds discussed in Trottet et al., 2022. Where appropriate, the rupture speed is corrected for finite slab thickness and bending stiffness to account for slab flexure effects. The rupture front is discretized into a sequence of subfaults: each subfault is activated at an onset time

$$t_o = \frac{x - x_{\text{start}}}{v_{\text{rupture}}},$$

using the subfault horizontal position x, the rupture starting position x_{start}, and the prescribed rupture speed v_{rupture}.

Each activated subfault is represented as a point/source in the SALVUS source array. The scalar seismic moment for a subfault is computed from the shear modulus, the prescribed slip, and the subfault width Δx:

$$M_0 = \mu\, x_{\text{slip}}\, \Delta x.$$

Moment-tensor components are set according to the user-specified mode-mix ratio, e.g. here the implementation uses $m_{xx} = 0,\ m_{xy} = M_0\, r_{\text{mode}},\ m_{yy} = -(1 - r_{\text{mode}})M_0$, which yields pure mode behaviour at the extremes of $r_{\text{mode}} = 0$ or 1, which means that an additional component of the moment tensor source becomes 0. All subfault sources use the same central frequency, wavelet shape, and amplitude scaling so the rupture is represented as a coherent moving source in the wave propagation simulations.

The following numerical settings and checks are applied consistently across simulations to ensure stability and reproducibility:

- Mesh resolution: element/subfault spacing h is chosen to resolve the cohesive zone and the shortest seismic wavelength; practical rule: $h \leq \min(l_{cz}/4,\ \lambda_{\min}/8)$, where $\lambda_{\min} = v_{\min}/f_{95}$ and f_{95} is the 95%ile source frequency.
- Time stepping: explicit scheme with CFL-like stability criterion $\Delta t \lesssim s\, h/v_{\max}$ and safety factor $s \approx 0.3$–0.5.
- Boundaries: use absorbing boundary conditions to minimize reflections at domain edges.
- Rupture discretization: subfault onset times follow $t_o = (x - x_{\text{start}})/v_{\text{rupture}}$; rise time is taken as $t_{rise} \approx l_{cz}/v_{\text{rupture}}$; scalar moment per subfault $M_0 = \mu\, x_{\text{slip}}\, \Delta x$.

Key parameters and their provenance used in the simulations are summarised below:

Parameter	Typical value / source
σ_n	0.00184 MPa (laboratory values cited in text)
τ_s	0.004 MPa (laboratory values cited in text)
G_I, G_{II}	From Mahajan and Joshi, 2008 (used to compute $\delta_{n,c}, \delta_{s,c}$)
v_s, v_p, ρ	From Guillemot et al., 2021 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)
l_{cz}	$l_{cz} = \frac{\pi}{8} l_{ch}$ with $l_{ch} = \dfrac{E_{\text{eff}} G_{II}}{\tau_s^2}$

4 Crack Propagation & Representation

4.1 Model implemented by me - Description and theoretical backround

- Cohesive zone model according to Mahajan and Joshi, 2008 (if not stated otherwise, this is the source for all bullet points below)
- Faulure either in weak layer or at interface where contacts between grains are bonded poorly (according to McClung, 1981; Schweizer et al., 2003)
- Cohesive zone model is different from previous models, because: it is assumed to have a narrow band fracture zone precedes the crack front. This means that before a point in the material shows cracking, there is microcracks or other weakening in the material
- In the model, this relation is described by a softening-force relation. This is different from linear elastic behavior (where the area under the force-displacement curve represents the separation of crack faces) because additional parameters are introduced: characteristic opening displacement and peak stress. Peak stress is the stress at which damage is initiated -> new crack faces are formed, traction between them is reduced to zero.
- Stress state in the snowpack due to uniform applied pressure is given by the lagrangian formulation for virtual work for dynamics:

$$\int_v s : \delta F \, dv - \int_{s_{int}} T \cdot \delta\Delta \, dS - \int_v \rho g \, \delta u \, dv = \int_{s_{ext}} T \cdot \delta u \, dS - \int_v \rho \frac{\delta^2 u}{\delta t^2} \cdot \delta u \, dv$$

- In this relation, F is the deformation gradient, V the volume in refrence configuriation (initial configuration), Δ the virtual jump displacement across the cohesive element faces, s_{ext} is the external surface area, on which the traction vector T with pressure and friction components is applied. S_{int} is the internal surface area of the body, ρ is the density and g gravitational acceleration
- Layers are assumed to be isotropic elastic: have young's modulus E and v, which is the Poisson's ratio
- Constitutive law for zero thickness cohesive elements at the interface (elements between boundaries): characterized by the stress-opening displacement potential (ϕ) function:

$$\phi(\Delta) = \phi_n \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left\{ \left[1 + \frac{\Delta_n}{\delta_n}\right] q \left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right] \right\}$$

- ϕ_n is the work of normal separation , ϕ_t of tangential separation, $q = \frac{\phi_n}{\phi_t}$ is the ratio between them. δ_n and δ_t are the characteristic normal and tangential separations as a function of traction.

- As the cohesive surfaces seperate, traction first increases to a maximum and then goes to zero with increase in separation. The cohesive surface tractions are written as:

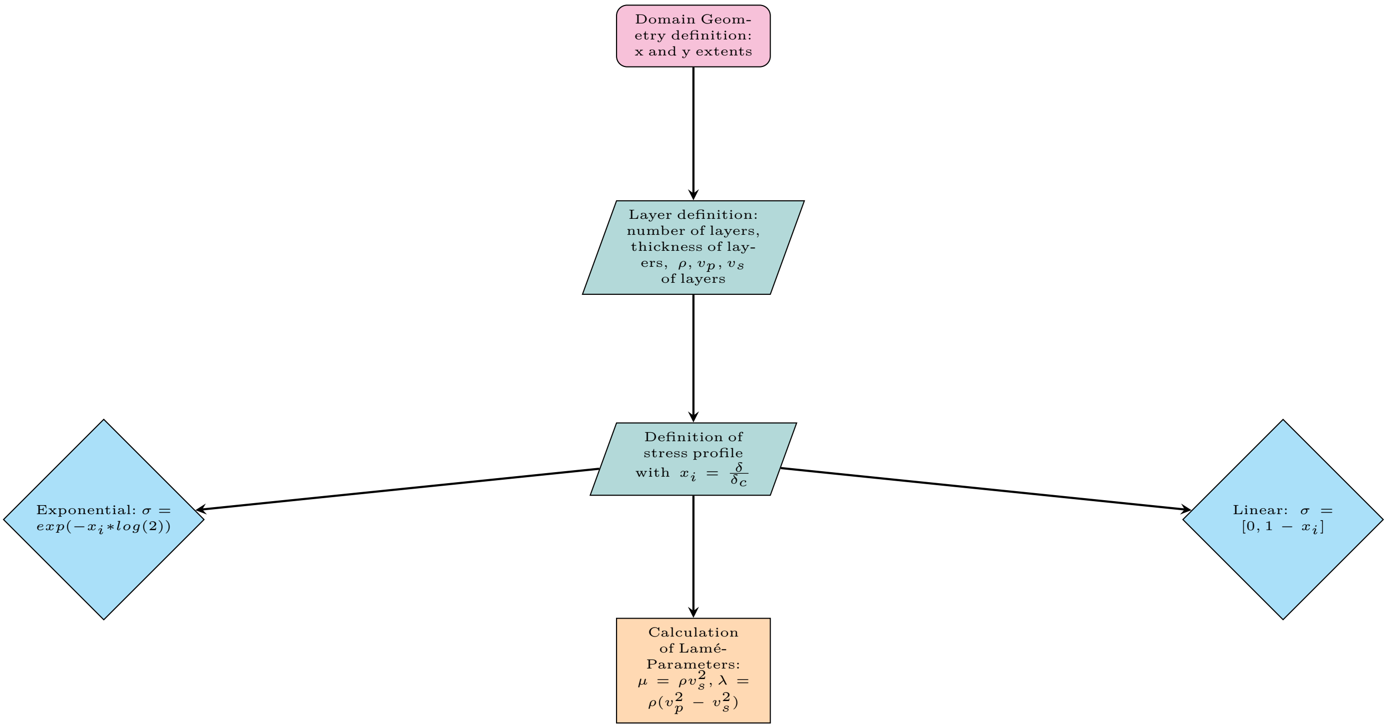
$$T_n = -\frac{\phi_n}{\delta_n} \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left\{ \frac{\Delta_n}{\delta_n} \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right) + (1-q) \left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right] \frac{\Delta_n}{\delta_n} \right\}$$
$$T_t = -\frac{\phi_n}{\delta_n} \left(2\frac{\delta_n}{\delta_t}\right) \left\{ q \left(1 + \frac{\Delta_n}{\delta_n}\right) \right\} x \exp\left(-\frac{\Delta_n}{\delta_n}\right) \exp\left(-\frac{\Delta_t^2}{\delta_t}\right)$$

- The normal traction is maximal when $\sigma_{max}, \Delta_n = \delta_n$, the shear traction is maximal when $|\Delta_t| = \frac{\delta_t}{\sqrt{2}}$
- The normal and shear work of separation are: $\phi_n = e\sigma_{max}\delta_n$ and $\phi_t = \sqrt{\frac{e}{2}}\tau_{max}\delta_t$, where e is exp(1), σ_{max} is the cohesive surface normal strength, τ_{max} is the cohesive surface tangential strength, δ_n, δ_t are the opening and shear distances which govern the fracture process ahead of the crack tip, when these have been exceeded, strain softening occurs
- The fracture zone is the distance between the crack tip and a point where one of these distances has been exceeded
- The parameters used come from laboratory measurements of snow
- In the model, Young's modulus (E) and shear modulus (μ) are calculated from shear and p wave speeds using standard elastic relations: $\mu = \rho v_s^2$ and $E = \mu\frac{3v_p^2 - 4v_s^2}{v_p^2 - v_s^2}$. This is done to convert measured values (wave speeds) into material properties
- Normal and shear fracture energy also from measured values in paper
- Function that calculates slip distribution: how much each of the subfaults slips. Cohesive weakening curve is integrated over the full normalized opening range (from 0 to 1). For each x position, the distance behind the crack tip is calculated, if the point is ahead of the tip, the slip is set to zero, if the point is in the cohesive zone, a normalized opening fraction is calculated from the distance to the tip and the length of the cohesive zone. The stress profile up to the opening is sampled and the released traction deficit integrated. If the point is far behind the tip, full slip is applied.
- The cracks are then written at defined x positions (at a depth with certain spacing). The cracks have a rise time, which is calculated according to the damage zone and is the time it takes one subfault to complete it's entire process. The onset time of each subfault is how long after the initial crack it fires (how long it takes the damage or the crack to reach this specific fault lodation), it's calcuated from the distance between the crack and the initial crack and the rupture speed.
- The slip of each subfault is dependent on it's location from the initial crack: it ranges from 0 to 1: $slip = \Delta_c\sqrt{4\xi(1-\xi)}$, where ξ is the parameter used for normalization so the last subfault reaches exaclty critical slip distance. The moment tensor forces are then calculated according to the slip for each location as well as the fraction between mode I and mode II opening
- Critical slip distance calculated from softening law: in linear softening law, the peak strength is the material tensile or shea strength, this is where breaking happens. At the point where the material fully fails, the stress drops to zero, this is the critical slip distance. Because the curve in the linear softening law is a triangle, the area of which is the fracture energy: $\delta_c = \frac{2G_c}{\tau_s}$, where τ_s is the shear strength of the interface. When this decreases, the critical slip increases.
- The values used for the simulation are: $\sigma_n = 0.00184MPa$, which is the normal interface strength, the shear interface strength $\tau_s = 0.004MPa$, the fracture energy of normal and shear modes were calculated from the total fracture energy and the mode mix. These come from lab experiments done
- The rupture speeds are calculated from the shear wave speed of the slowest layer. This is adjusted for the stiffness and the thickness of the slab: this is done because stiffer slabs have more slab bending, thicker slabs have more gravitaional energy (slab pull). The numerical values used are typical snow values. The rupture speeds are then calculated from the bounds defined by Trottet et al., 2022: 0.25 to 0.6 times the shear wave speed for sub-rayleigh, and 1.6 times the shear wave speed for supershear.

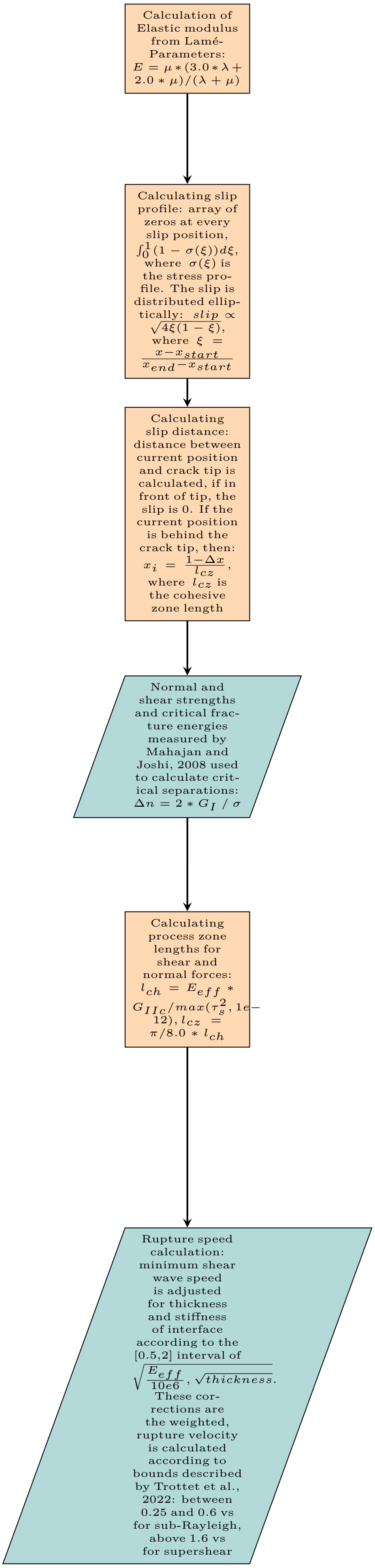
- The characteristic length of a crack is calculated using the Irwing characteristic length: $l_{ch} = \frac{E_{eff} G_{II_c}}{\tau_s^2}$, the process zone length using Dugdale-Barenblatt approximation: $l_{cz} = \frac{\pi}{8} l_{ch}$

4.2 Code workflow

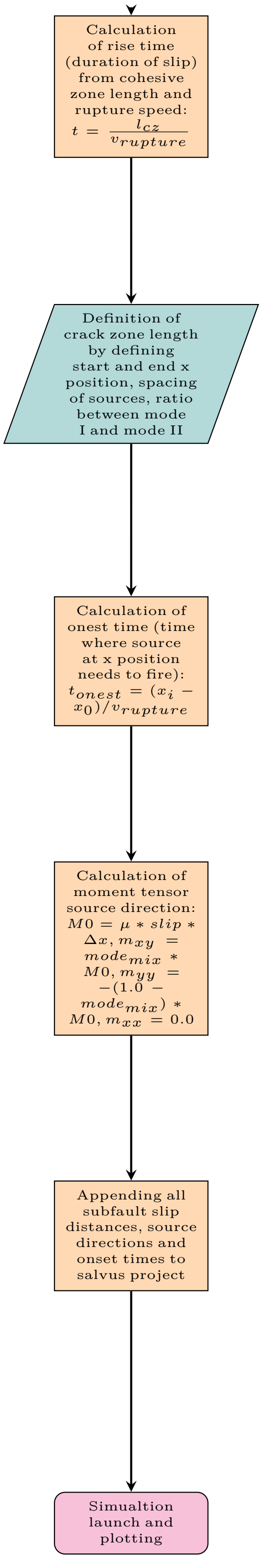
Part 1: Domain and Stress Profile Setup



Part 2: Material Properties and Process Zone Calculation



Part 3: Rupture Propagation and Source Definition



4.3 Crack Types Relevant for Avalanche Release

- Anticracks: Cracks that close after opening (Gaume et al., 2018), in other words, this means that the weak layer collapses under loading, which leads to the opening of cracks in layers above because of the collapse (Knight & Knight, 1972). This introduces frictional contact between the crack faces.

- Mode I: space is opened between the crack faces under tensile loading (Heierli et al., 2003).
- Mode II: parallel shearing (Heierli et al., 2003).
- Mode III: diagonal shearing (Heierli et al., 2003). (see figure in Mahajan and Joshi, 2008 for clarifiction)

4.4 Terminology according to Ramos et al., 2022

- Cohesive zone Λ: fundamenta length scale in rupture models where the fault strength decreases from static to dynamic level
- Critical slip distance: slip needed for fault strength to drop from static level to dynamic level
- Cracklike rupture: in such a model, the rise time is about equal to the total rupture duration: this means it takes the same amount of time for the slip to reach its highest value, as the time the whole fault rupture lasts
- Dynamic fault strength: fault strength during the rupture: $\sigma_d = \sigma_n^{eff} x \mu_{dynamic}$, which is the product of effective normal stress and the dynamic friction coefficient
- Dynamic stress drop: difference in shear stress before and during earthquake (rupture)
- Fracture energy: energy needed to grow a propagating crack
- Slip: displacement at a given location on a fault
- Stress drop: difference between static strength (strength before) and thedynamic strength (during rupture)
- Static stress drop: difference in shear stress before and after, the average over the fault area is $\Delta\sigma = \bar{\Delta\sigma_s} = \frac{1}{A}\int \Delta\sigma_s dA$, where σ_s is the static fault strength
- Static afult strength: fault strength before movement: $\sigma_s = \sigma_n^{eff} x \mu_{static}$

4.5 Other points from Ramos et al., 2022

- In mode II ruptures, two degrees of freedom lead to the generation of P- and SV-waves (vertically polarized shear wave)
- Mode III can only generate SH-waves in homogenous media
- Mode I represents tensile crack which is not usually important for dynamic rupture, but point-source models can account for fault opening by separating tensor into different components
- Different modeling methods use different forms of eqations: FE and SE use weak form, PSE and FD use strong form. Weak form implicitly satisfies Neumann boundary conditions, which means that only Dirichlet need to be implemented. Meshes can also be more complex, because a lower oder derivative of the equations is used
- During dynamic rupture, the cohesive zone width is a critical parameter, it can be estimated depending on the friction law used: for the linear slip-weakening friction law (frictional coefficient decreases linearly as slip propagates): $\Lambda = \frac{C_1}{C_2}[\frac{GD_c}{\Delta\sigma_d}]^2[\frac{1}{1+\frac{L_0^2}{L^2}}]$, where L_0 is the critical half-crack length
- Friction law describes how the friction evolves over time, this is an important choice
- Slip weakening when slip is greater than dynamic friction coefficient

- SLip strengthening when slip is smaller than static friction coefficient -> this can be used to model crac k arrest
- No energy to grow crack when they are equal
- Velocity dependent frcition weakening law

4.6 Processes linked to crack propagation

- Loading & loading rate: initial loading which can trigger crack propagation. The propagation of cracks can be upslope from the loading point, which can mean that avalanches are triggered remotely (Gaume et al., 2018) or propagation can be downslope, which means that the avalanche is triggered at the loading point (Gaume et al., 2018). Which direction the propagating cracks have, is influenced by model scale, as discussed by Gaume et al., 2018. Loading can be slow, for example by snow accumulation, or rapid, for example by a skier or an explosion. Slow loading can lead to sintering, which means that new bonds are formed between snow grains, which can strengthen the material and delay failure (Mulak & Gaume, 2019). Rapid loading does not allow for sintering to occur, which means that there is still failure (Mulak & Gaume, 2019).
- Coupled criteria: There needs to be enough stress on the slab to exceed the weak layer strength (enough force for cracks to initiate because of failure), but also enough energy so that cracks can propagate. This means that two conditions need to be met for avalanche release.
- Snow Mechanical Properties: properties of the snowpack influence crack propagation. Slab bending in steep terrains for example can lead to supershear crack propagation (Trottet et al., 2022). The weak layer thickness also has an influence on crack propagation (Gaume et al., 2015).
- Strain softening: this means that less stress is required to have deformation, in other words, the material becomes weaker as it deforms. This is important because strain softening can lead to faster crack propagation, which could lead to faster slab breakoff.

4.7 Crack Areas

- Crack tip: front of the crack, this is where the crack propagates
-

4.8 Governing equations of crack propagation

- Nucleation: Stress condition: $\sigma\sigma_c$, which means that the stress due to loading (σ) must exceed the weak layer strength (σ_c) over a certain distance (Rosendahl & Weißgraeber, 2020).
- Energy criterion: GG_c: the energy available from the slab deformation (G) needs to be greater or equal than the fracture toughness of the weak layer, which means there needs to be enough energy for new cracks to form and to propagate (Rosendahl & Weißgraeber, 2020).
- Critical crack length: $r_c = \Lambda\sqrt{\frac{2E'G_c}{\sigma^2}}$. E' is the slab elastic modulus, $\sigma$the applied stress. The critcial crack length is the length at which a crack becomes self-propagating and can lead to slab breakoff (Gaume et al., 2015).
- Timoshenko beam representation: takes into account shear and rotational deformation of a beam, Mahajan and Joshi, 2008 represent the snow slab as a Timoshenko beam.
- Slip and displacement: relative displacement between slab and weak layer (Mahajan & Joshi, 2008)
- Fracture energy: Energy required to create a crack is the area under the traction-separation curve (Mahajan & Joshi, 2008)

4.9 Crack representation in past models

4.9.1 Burridge, 1973

- discusses different cases for crack propagation speed based on equations
- Modeled a crack that propagates at a constant velocity from a point
- Crack modeled has friction, but no cohesion, this means that the resistance which needs to be overcome is only from friction
- For rayleigh-speed propagating cracks: a peak shear stress develops ahead of the crack tip
- Speed of the crack depends on static friction: if this is high, then crack propagates sub-Rayleigh
- If static friction is lower, then the low stress peak at the front can exceed material strength before the crack arrives

4.9.2 Andrews, 1976

- Stress drops when slip starts: this is mathematically the same as motion
- Material can only support a certain amount of stress: stress singularity satisfying fracture criterion (critical stress needed for fracture formation)
- Elastic energy loss at the crack tip in previous media
- Different friction law where friction at crack tip decreases linearly over critical slip distance
- Mother daughter mechanism of cracks: crack nucleates at sub-rayleigh speed, stress peak grows at it accelerates toward rayleigh speed, stress peak nucleates a daughter crack, which moves at supershear speeds, eventually the daughter crack fuses with the mother crack again

4.9.3 McClung, 1981

- linear elastic fracture mechanism: stress is singular at the crack tip –> only works if there is no deformation before crack propagation
- Discusses a shear crack model: shear band & slip deformations
- Displacement of crack is related to stress applied
- No friction on crack faces
- End zone is the length where the stress drops from initial stress to the residual stress. This is observed because of strain softening: stress needed for same displacement is less with time
- If crack length is about the same as the size of the zone where deformation is plastic, there is no propagation becaus ethe energy needed for propagation is too large and there is yielding instead of propagation of cracks
- The propagation of cracks is dependent on the intensity factor of modes I and II stresses, initial and residual stress, the average slip in the end zone. Resulting from this is the propagation condition for the shear band.
- The crack length might be many slab thicknesses large, this means that there is a relationship between slab thickness
- Model focuses on large-scale deformations instead of microscale failure. These are independent of slab properties

- Model represents crack propagation in weak layer (described as shear band) with initial loading, but no loading after initial loading
- Shear band is described by contrast in material properties
- Different shapes of weak layers discussed: band, but also elipsoidal inclusion with fixed location
- For elipsoidal zone: stress is related to normal strain, weakened zone parameters, shear stress magnitude and axis ratio of elipsoidal weak zone
- Incremental relationships describing yielding vs dilatancy (volume increase under loading), which can also be made slope-angle dependent
- Plastic softening / hardening is 0 in a flat zone
- In this model, slope angle has no effect on critical moduli relations, but it's important for plastic deformation
- There can be localized shear band formation (new shear bands form in different locations) because deformation pattern can bifurcate
- The first fracture observed is a tension fracture whcih propagates across the slope because of stress propagation in the weak layer. The angle between the shear bed and the tension fracture is observed to be $90° \pm 10°$.
- Tension stresses cannot be formed before there is a shear band in this model (no stresses without a weak layer)
- Stresses at different locations of shear band can be calculated
- If distance needed for stress to drop from initial to residual is lower than one slab thickness, then the principal tension stress increases and concentrates -> again slab thickness dependency
- If the propagation speed increases, the initial angle of propagation also increases
- Shear disturbances need some anisotropy to propagate
- Initial fracture location dependent on stress concentration due to topographic effects
- Sequence of fractures: Initial shear failure first, the strain softeing, propagation of the shear disturbance, followed by tension fracture in the snow slab which is concentrated at the shear bed surface, loading drives disturbance further, which leads to flank fracturesa and lastly failure at the lower boundary of slab
- Weak layer thickness has no effect in this model because deformation is mostly shear

4.9.4 Bazant and Planas, 1998

- Fracture mechanics determines material failure by energy criteria, sometimes in combination with strength criteria
- Fracture theory considers failure to propagate through the medium rather than through the entire failure zone at the same time
- Previously, fracture mechanics only applicable to homogenous britte materials (like glass, which is homogenous but still fails in a brittle manner)
- This is a problem, because many materials, including snow, are heterogenous but still have brittle failure: this book focuses on concrete
- Linear elastic failure mechanics based on the solution for stresses at the vertex of an ellipsoidal cavity: stress cannot be used as a criterion because at the crack tip, stress is infinite –> crack propagation if energy is available

- Energy which needs to be available is $2\gamma_s$, where γ_s is the specific surface energy of the elastic solid
- Energy required for crack propagation is larger than specific surface energy, it was later defined as crack resistance -> can be described as a curve dependent on crack length
- Fracture mechanics requires a certain energy to form crack: crack initiation energy
- Smeared cracking: does not take into account fracture mechanics, stress is a finite element limited by tensile stregth of material. after stress limit is reached, stress must decrease -> gradual release of stress as strain softening
- Two main types of failure: plastic and brittle. In plastic failure, structure develops a single degree of failure mechanism, this means that failure proceeds simulatiously at different points in the material. Plastic failure when the yield plateau is long on the load-deflection diagram: large area where deformation is plastic. If such a plateau does not exist, there is brittle failure

4.9.5 Mahajan and Joshi, 2008

- Desrcibe a cohesvie zone model (CZM)
- Focuses primarily on mode II fractures
- Model is defined by 4 physical properties of snow, which were determined by experments: tenisle and shear stregth, normal and shear fracture energy
- Here, a narrow-band fracture can exist ahead of the crack tip. This is important because the fracture process is distributed over a finite zone, so the crack can be described by a traction-displacement relation across the band rather than by a single sharp tip.
- In the CZM, there is softening to force relation: the opening of the cracks and peak stresses are contained under the force-displacement curve -> softening law
- Cracks occur, when the peak stress occurs and traction is reduced to zero
- Material characterized on both sides of narrow-band frcature: relation between traction and displacement across the band
- Initial crack needed for further failure in model
- Mass matrix as a combination of internal and external forces
- Initial forces contribute to cohesive element tracttions, external forces to body forces and external loads
- Stress opening is a potential function with tangential and normal components, which have characteristic normal and shear separation distances
- Traction decreases as cohesive surfaces separate, it approaches 0 as the separation increases
- Strain softening occurs in this model when the separation distances corresponding to cohesive normal and tangential strength have been exceeded

5 Notes

- Look into wave modes (flexural and other mode)

6 Crack Propagation

6.1 Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments (Bergfeld et al., 2025)

Main Question of article: Can supershear crack propagation be observed in snow avalanche field experiments and are there different stages of crack propagation?

Why this is relevant for the thesis: Possible different stages of crack propagation speeds and their transitions are important for DAS data interpretation so that accurate risk assessments can be made.

Previous to Bergfeld et al., 2025, supershear fractures in snow have been observed in many numerical models, but never in slope experiments. Bergfeld et al., 2025 conducted controlled experiments on a snow-covered slope and did optical tracking of cracks, they also used MPM modeling to validate the slope experiments. Both the experiments and the numerical models showed that there are different stages in crack propagation (Bergfeld et al., 2025). Understanding the different stages of crack propagation speeds and what causes the transitions between them is important to make accurate risk assessments based on DAS data: do cracks propagate slowly which causes a slow breakoff of the slab or is the propagation supershear? The different stages of crack propagation observed by Bergfeld et al., 2025 correspond with different locations in the snow: at the lower end of the snowpack, the deformation was dominated by slab bending induced by weak-layer collapse, whereas at the higher end, the deformation was dominated by slab tension which was induced by the shear failure of the weak layer (Bergfeld et al., 2025). The results shown by Bergfeld et al., 2025 show that anticrack propagation dominates the early stage of crack propagation, but after a certain propagation distance, shear-driven fractures occur, especially on steeper slopes. This means that crack propagation leading to avalanches is a combination of both anticrack and shear fractures (Bergfeld et al., 2025). Furthermore, in supershear crack propagation in the simulations by Bergfeld et al., 2025, crack propagation was not arrested by slab fractures. This means that the release areas are much larger, because there is more extensive fracturing and faster propagation (Bergfeld et al., 2025). For sub-Rayleigh crack propagation, Bergfeld et al., 2025 observed crack arrest if there are existing slab fractures, which leads to smaller release areas. This is critical for the thesis because if crack propagation is slower, this means that the avalanche released is likely smaller. This is only a valid conclusion if the individual cracks can be observed

6.2 Numerical investigation of crack propagation regimes in snow fracture experiments (Bobillier et al., 2024)

Main Question of article: How do different snow mechanical parameters influence crack propagation speed?

Why this is relevant for the thesis: Understanding the influence between crack propagation speed and snow parameters is important to understand when cracks can be observed in DAS data if at all and how long it takes between initial cracking and avalanche release.

In previous models of crack propagation in snow, the mechanical properties of snow were not considered (Bobillier et al., 2024). A 3D discrete element model was developed by Bobillier et al., 2024 to investigate the mechanical properties of snow of crack propagation in a PST, this was done at different slope angles. Bobillier et al., 2024 assumed in their model that sintering and other time-dependent effects have no influence on crack propagation. A three-layered model consisting of a rigid bottom layer, a weak layer resembling depth hoar, as well as a dense slab layer was implemented (Bobillier et al., 2024). This type of model can possibly also be considered in the thesis. Understanding how different mechanical parameters affect crack propagation is relevant for the thesis because it influences if cracks can be observed in the DAS data at all, and if they are, how long it takes between initial cracking and slab breakoff, which is dependent on crack propagation speed. During the propagation saw test, the crack propagation speed reaches steady state, the numerical simulation results by Bobillier et al., 2024 show that this is largely dependent on slope angle. At slope angles lower than 30°, the steady-state propagation speed is sub-Rayleigh, if the slope angle is greater or equal to 30°, the steady-state speed is larger than the shear wave speed, so supershear (Bobillier et al., 2024). It also depends on the slope angle how long it takes to reach this steady-state speed (Bobillier et al., 2024). Bobillier et al., 2024 show that the higher the slab elastic modulus, the lower the normalized crack propagation speed, this is the opposite for weak layer elastic modulus: the higher the weak layer elastic modulus, the higher the crack propagation speed. If the weak layer is very weak, crack propagation speed is also very low, Bobillier et al., 2024 explain this by failure occurring before cracks have propagated. Direct failure also occurs when the slab is very soft (Bobillier et al., 2024). These considerations are important for the thesis because it can mean that cracking might not be observed before the slab break off, or if the system is too stable, cracking is observed but there is no avalanche.

6.3 Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches (Trottet et al., 2022)

Main Question of article: What are the factors which influence cracks to propagate at speed faster than shear wave speed in snow slabs and how can this be modeled?

Why this is relevant for the thesis: Understanding at which speeds cracks propagate under different conditions and whether the cracks cause avalanches is important to make accurate risk assessments based on DAS data.

Previous to Trottet et al., 2022, two main theories about the propagation of cracks in snow slabs have been discussed. Some authors investigate shear only modes of failure, while others investigate anticrack modes (Trottet et al., 2022). In nature, large crack speeds higher than the velocity of shear waves were observed, while in smaller-scale experiments, sub-Rayleigh propagation was observed (Trottet et al., 2022). Understanding at which speeds cracks propagate at which modeling scales is important to accurately assess the risk associated with crack observations in DAS data: if cracks propagate at high speeds, slab breakoff will occur faster than if cracks propagate at lower speeds. Trottet et al., 2022 conducted multi point material modeling of propagation saw tests at different angles. Trottet et al., 2022 observed that the onset of supershear fractures depends mainly on terrain angle, on flatter terrain, the stress induced by slab bending causes propagation upslope, whereas in steep terrain, propagation was observed downslope because the normal stress is higher than shear stress. In inclinations which are lower than the effective friction of the material, the propagation speed was sub-Rayleigh, whereas at angles higher than the effective friction, supershear propagation was observed (Trottet et al., 2022). This was explained by Trottet et al., 2022 by slab bending. In flatter terrain, the bending of the slab is limited by the collapse depth (the slab can only bend as much as the snow layer collapses because the terrain is flat), which leads to a constant speed of crack propagation (Trottet et al., 2022). In inclined terrains, slab tension can increase, which also leads to crack propagation speeds to increase (Trottet et al., 2022). This is also an important consideration for DAS observations to assess risk correctly because of the time passed between initial crack and the breakoff of the slab. The slab considered in the simulations by Trottet et al., 2022 is pure elastic, at lower tensile strength of the snow this leads to initial cracking at the surface, crack arrest because of en echelon cracks and consequently smaller release areas. At higher tensile strengths in slabs, fractures occur at the bottom of the slab, the cracking does not influence crack propagation as much and release areas are larger (Trottet et al., 2022). Trottet et al., 2022 come to the conclusion that large slab avalanches with hard slabs (high tensile strength) show supershear cracks, they also mention that small-scale crack experiments are not representative of actual speeds.

6.4 Modeling snow slab avalanches caused by weak-layer failure - Part 2: Coupled mixed-mode criterion for skier-triggered anticracks (Rosendahl & Weißgraeber, 2020)

Main Question of article: How can anticracks in snow be described by different stress conditions (skier triggered) and how does critical stress needed for slab breakoff change with slope and snow parameters?

Why this is relevant for the thesis: Understanding what causes critical cracks to form is understand to predict where they can be observed in DAS data

Rosendahl and Weißgraeber, 2020 describe that classical models only account for growth of existing cracks, caused by instabilities in weak layers. The new model developed by Rosendahl and Weißgraeber, 2020 does not assume existing weak layers, here, a slab release only occurs if there is enough energy for the crack to propagate and the critical stress is reached. These considerations are important for DAS data observations, because the snow stratigraphy might not be known in the area, which would mean that assumptions about existing weak layers cannot be made. Rosendahl and Weißgraeber, 2020 distinguish between two failure modes: Mode I, where cracks open normal to crack faces, and mode II where cracks open tangential to existing crack faces, the model developed accounts for both failure modes by considering the ratio between the shear and compressive force. Independent of the shear and compressive force ratio, Rosendahl and Weißgraeber, 2020 observe that the critical skier force needed for failure decreases with increasing slope angle. This means that in very steep terrain, an avalanche can also release without outside forces (Rosendahl & Weißgraeber, 2020). The numerical modeling results show that as the slab thickness increases, the critical skier force also increases, Rosendahl and Weißgraeber, 2020 explain this by the load distribution. Thicker slabs can distribute skier load more evenly, this is only true up until a certain thickness however, where the slab becomes too heavy (Rosendahl & Weißgraeber, 2020). The same is true for Young's modulus: the higher the Young's modulus of the slab, the stiffer and more resistant to deformation by the skier it is (Rosendahl & Weißgraeber, 2020). Rosendahl and Weißgraeber, 2020 explain this by lower Young's moduli slabs deforming more easily, which leads to more bending of the slab, which results in the load being concentrated on one part of the weak layer, which leads to easier failure. The new model developed by Rosendahl and Weißgraeber, 2020 observes that failure mode II dominates in more steep terrain, this observation is important when considering DAS data, because there might be directional effects between the orientation of the cracks and the cable in the data. The major limitation of the model, as stated by Rosendahl and Weißgraeber, 2020 is that the model relies on linear elastic solutions. This means that time-dependent processes, such as sintering are not taken into account

6.5 Investigating the release and flow of snow avalanches at the slope-scale using a unified model based on the material point method (Gaume et al., 2019)

Main Question of article: How can crack propagation and slab breakoff be modeled using the material point method in 2D and 3D?

Why this is relevant for the thesis: The material point method will also be used in the thesis, understanding how to model slab breakoff with this and which considerations need to be taken is important. Furthermore, it is important to understand how cracks propagate and how this can be modeled to understand and model DAS data of avalanches.

Gaume et al., 2019 state that snow can behave like a solid or like a fluid depending on the type of deformation which is present, modeling this behavior is important to understand which types of loading correspond to which behavior and deformation. The release of avalanches is because of failure initiation in a weak layer, the onset of crack propagation in the weak layer and the resulting detachment of the sliding slab (Gaume et al., 2019). Because the weak layer has porous character, there is volumetric collapse, which leads to the closure of cracks in the weak layer, so called anticrack behavior (Gaume et al., 2019). The new approach to model avalanche dynamics described by Gaume et al., 2019 uses the multi-point method to model deformation. Here, particles in the snowpack are used to track mass, momentum and deformation of the snowpack (Gaume et al., 2019). This means that individual particles instead of fixed mesh points are used for modeling, the lack of connectivity between the points complicates the calculation of spatial derivatives, so an Eulerian background is combined with interpolation functions by Gaume et al., 2019 to circumvent this problem. This is also an important consideration in case the MPM method will be used in the thesis. The model described by Gaume et al., 2019 uses a mixed-mode shear and compression yield surface and accounts for softening and hardening of the weak layer. The model accounts for the fact that the internal friction of snow is usually lower than the dynamic friction in the cracks, these are described by different parameters in the model (Gaume et al., 2019). The simulations run by Gaume et al., 2019 include slope-scale simulations in 2D and 3D with a constant slab depth, which account for spatial variability of the snow depth. The failure in the weak layer is initiated by loading at the bottom of the slope (Gaume et al., 2019). The failure propagates along the weak layer in a mixed-mode anticrack, the propagation speed modeled by Gaume et al., 2019 shows large variations. Changes in slope angle and curvature for example have large influences in the speed: concave leads to high propagation speed, whereas convex parts of the slope have lower speeds (Gaume et al., 2019). The propagation speed of the failure cracks is also influenced by slab tensile fractures: the closer to a slab tensile fracture, the lower the propagation speed (Gaume et al., 2019). The propagation speed of the failure cracks observed by Gaume et al., 2019 also showed strong directionality: propagation in the slope direction (uphill) was observed to be 2x higher than in the downslope direction. Understanding the variations in propagation speed is very important to make sense of DAS observations, for example the time passed between loading and slab breakoff and how this can be explained. The slab is released where the slope angle exceeds the friction angle of the material (Gaume et al., 2019). This is an important consideration when thinking about the observation of slab breakoff in DAS data, because it can be assessed where the breakoff will happen: does it happen far away from the cable when signal could be attenuated or not? Gaume et al., 2019 state that the initial crown fracture of the slab breakoff is followed by a staunchwall and flank shear fractures. Could all of these different fractures be seen in DAS or are they maybe recorded as one event? Broken slab pieces can stick together again, the collective behavior of these particles leads to fluid-like behavior (Gaume et al., 2019). For remote triggering of avalanches, there needs to be the collapse of the weak layer as well as fracture propagation (Gaume et al., 2019). The collapse of the weak layer is negligible on steep slopes according to Gaume et al., 2019, the large angle is sufficient to observe failure by shearing alone. Here, weak layer collapse is secondary to failure Gaume et al., 2019. Could these separate processes also be observed? Like weak layer collapse would lead to deformation of DAS cable, which could be an indicator even before slab breakoff is observed, but of course only for flatter slopes. Gaume et al., 2019 observed a top to bottom collapse (from top of hill to bottom) in their simulations, which is consistent with large-scale PST experiments. The limitations of the model described by Gaume et al., 2019 is that the strength of the weak layer is not dependent on strain rate. This means that there is no potential for sintering as described by Mulak and Gaume, 2019, the model by Gaume et al., 2019 can therefore only be applied for fast loading, for example for failure induced by a skier. Another limitation of the model is that there is only variation in snow depth, other variations in other parameters are not accounted for by the Gaume et al., 2019 model.

6.6 Dynamic anticrack propagation in snow (Gaume et al., 2018)

Main Question of article: How can anticrack propagation in snow be modeled in a way which also represents strain softening accurately?

Why this is relevant for the thesis: Understanding how strain softening and crack propagation is modeled is important to make accurate models of DAS data

Models previous to Gaume et al., 2018 accounted for anticrack propagation by removing mesh elements. Gaume et al., 2018 developed a new model which accounts for the closure of cracks during anticrack propagation. Understanding this type of modeling is important for the thesis because anticracks are believed to be the main cause of snow slab avalanches. Modeling the correct type of material behavior and crack propagation is important to model DAS data accurately. Gaume et al., 2018 modeled hardening and softening of the slab by a yield surface with varying pressure, this leads the model to include strain-softening even at macroscopic uniaxial compression. Understanding snow material and model response is important to make sense of DAS data: which stages of deformation or material response can be identified? The results of the simulations show stages in material response: the first stage is an elastic regime, followed by a pressure and shear stress drop, then a volumetric collapse indicated by plastic consolidation, with the last stage being dense packing (Gaume et al., 2018). The results show important distinctions of slope simulations and propagation saw tests: Gaume et al., 2018 show that for slope simulations, failure occurs bottom to top (like remote triggering), whereas for smaller scale PST simulations, failure occurs top to bottom. This is a crucial consideration for simulation of DAS data because it shows where failure can be observed in terms of cable location. Gaume et al., 2018 explain the different propagation behavior by the strain caused by slab bending: at slope scales, the tensile stress caused by slab bending is not large enough to cause fracture, whereas in PST it is. Again, a MPM was used in combination with an Eulerian mesh to make time-dependent calculations easier.

6.7 Snow fracture in relation to slab avalanche release: critical state for the onset of crack propagation (Gaume et al., 2017)

Main Question of article: How can the critical crack length needed for avalanche release be modeled and which parameters influence this?

Why this is relevant for the thesis: The critical crack length and the parameters which influences this are important because it influences what can be seen in DAS data. This is in relation where data is measured: slope angle etc has an influence on how many avalanches are recorded by DAS and subsequently how large the dataset for breakoff observation is

Gaume et al., 2017 develop an equation which describes the relation of critical crack length to snowpack parameters. The release of a dry-snow slab avalanche requires the formation of a weak layer, the initial failure is a macroscopic crack in the weak layer (Gaume et al., 2017). The length of this crack cannot exceed the critical crack length, because else there will be material failure and avalanche release (Gaume et al., 2017). This is because the critical crack length is the length of the crack which is present when the shear stress is equal to the shear strength of the material. Understanding the influences of different snow and terrain parameters on critical crack length is relevant for DAS observations of avalanche breakoff because it can be understood how many avalanches occur and why. Previous modeling of avalanche release to Gaume et al., 2017 made simplifications to crack propagation, which results in inaccurate predictions. Gaume et al., 2017 estimate critical crack length using a discrete element model as described by Cundall and Strack, 1979, where the cracking is due to sawing in the weak layer, similar to Gaume et al., 2015. The modeled data is then compared to experimental data. The numerical simulations by Gaume et al., 2017 show that critical crack length increases, if elastic modulus of the slab increases or the weak layer thickness increases. The critical crack length on the other hand decreases if the slab density increases, the slab thickness increases or the slope angle increases (Gaume et al., 2017), this means that under these conditions, cracks need to be less long for failure to occur. Gaume et al., 2017 state that the shear stress at the crack tip can be decomposed into a part caused by slab tension and a second part caused by slab bending. The analytical expression for critical crack length taking into account snow and terrain parameters agree with experimental results of saw tests conducted by Gaume et al., 2017. The length of the critical crack is also relevant for the thesis because maybe shorter cracks show up less well in DAS data (less of a displacement). The analytical expression derived describes critical crack length better than the anticrack model, especially in regard to weak layer thickness and slope angle (Gaume et al., 2017). The anticrack model shows that critical crack length is not dependent on slope angle, this is because the model makes assumptions for simplification which the new expression by Gaume et al., 2017 does not make. The new expression derived by Gaume et al., 2017 has some limitations, the dependence of critical length on slope angle could also be caused by young's modulus and slab thickness, if these two are decreased, the critical length decreases less with increasing slope angle. Furthermore, Gaume et al., 2017 do not account for the shear stresses caused by slab bending, but suggest that this effect is small because simulation results are in good agreement with observations. The models described here assume a homogenous slab, where in reality the snowpack has multiple layers of different strengths, which can have an effect (Gaume et al., 2017).

6.8 Modeling of crack propagation in weak snowpack layers using the discrete element method (Gaume et al., 2015)

Main Question of article: Which snow parameters influence crack propagation distance and speed and how can these processes be modeled?

Why this is relevant for the thesis: Understanding the influence of different snow parameters on crack propagation distance and speed is important to model and understand slab breakoff DAS data in thesis.

Gaume et al., 2015 state that dry-snow slab avalanches form by the failure of a weak snow layer underlying a cohesive slab. The local damage leading to failure is a crack in the weak layer, which when it reaches critical length leads to tensile fracture in the slab, which leads to avalanche release (Gaume et al., 2015). Understanding how these processes can be modeled and which parameters affect the propagation distance and speed is important for the thesis because slab breakoff is investigated in DAS data. Understanding which snow parameters lead to large crack propagation or speed is important to determine in which DAS datasets slab breakoff is most likely to be detected and at which times in the dataset this would be (does the breakoff take a lot of time or not because speeds and distances are low or high?). Gaume et al., 2015 state that the dynamic phase of crack propagation and the effects of different snow parameters are poorly understood, which limits avalanche forecasting because it is unclear how and under which circumstances cracks propagate. This is directly linked to the thesis, because cracking of the snow layer is investigated in DAS data. The method used by Gaume et al., 2015 to model cracking in snow is the discrete element method described by Cundall and Strack, 1979, with a limitation that the slab is modeled as a continuous material with elastic-brittle behavior. The loading modeled is due to gravity and the propagation saw test (Gaume et al., 2015), the cohesive part of the snow (so the connections between the grains) are modeled by the linear contact law, the maximum and minimum and minimum tensile and shear stresses are modeled by beam theory (Gaume et al., 2015). The stiffness at the bonded contact (of grains) is calculated (Gaume et al., 2015), this is important to see how the material reacts: what influence does the stiffness of the grain contacts have on crack propagation distance and speed? Gaume et al., 2015 calculate the time-step of the simulations from the snow grain properties, this ensures that the maximum crack propagation velocity is lower than the speed of elastic waves in the snow, this means that the cracks in the snow cannot propagate faster than the propagation of the elastic waves which are caused by the loading. Understanding the relationship between elastic wave velocity and crack propagation speed is also important for the modeling of DAS data of slab breakoff which will be done in the thesis. Gaume et al., 2015 determined the propagation distance of cracks in the slab by setting the tensile and strength to finite values, to determine propagation speed on the other hand, these values were set to infinity. Understanding the propagation speed and distance of cracks is important because the higher either of these values, the faster the avalanche breaks off. The propagation distance described by Gaume et al., 2015 is defined as the distance between the point where failure initiates, so the point where the initial load is applied and the point where the tensile failure is located. The numerical models showed that the crack propagation speed increases with increasing young's modulus, increasing thickness of slab and increasing density, but the thickness of weak layer has no influence on propagation speed (Gaume et al., 2015). From their results, Gaume et al., 2015 suggest that the lower the young's modulus of a snow layer, meaning the softer a snow layer, the lower the crack propagation velocity, which means that avalanches break off less fast. This suggests that the propagation speed of cracks is mainly influenced by failure conditions such as the tensile strength of the snow layer rather than structural parameters such as weak layer thickness (Gaume et al., 2015), which is relevant for the thesis because factors such as weak layer thickness need not be modeled, which saves computational time. The propagation distance of cracks is influenced by weak layer thickness however, Gaume et al., 2015 suggest that if weak layer thickness increases, there is more slab bending which leads to a higher tensile stress and more likely tensile failure, which means that crack propagation distance decreases, because failure happens before the cracks can propagate very far (Gaume et al., 2015). Furthermore, the simulations run by Gaume et al., 2015 show that the slope angle has little influence on crack propagation for low tensile strength snow layers, but for high tensile strength layers there is a large correlation between slope angle and propagation distance of cracks (Gaume et al., 2015). This can be explained by the main failure mode being tensile failure due to slab bending at high tensile stresses of the snow layer (Gaume et al., 2015). Gaume et al., 2015 conducted lab experiments, which confirm the results of the numerical simulations in all but one case: the relationship between high young's modulus and large crack propagation is counterintuitive, because in harder snow, cracks propagate less well compared to soft snow, Gaume et al., 2015 explain this by the limitations of the beam theory to account for dynamic effects. This is an important consideration depending on which method will be used to model crack propagation in the DAS data. Gaume et al., 2015 show that the stress evolution in the snowpack always shows an increase of tensile stress at the top of the slab before crack propagation, which happens close to the tip of the crack in the weak layer. The tensile crack opens up as soon as the tensile stress is equal to the tensile strength of the snow, this crack always starts at the top surface of the slab (Gaume et al., 2015). Understanding the stress evolution of the snowpack is important for DAS data interpretation, because it is possible that the different phases of this path (increase in stress, opening of crack in weak layer, tensile cracking and failure) can be seen in the data. Gaume et al., 2015 mention two important limitations of the simulation results: the models assume a simplified shape of the weak layer, as well as the assumption that the crack propagation speed is lower than the speed of the elastic wave. At high propagation speeds of the crack, the tangential displacement of the snowpack becomes more important than the vertical displacement (Gaume et al., 2015). Furthermore, the column length of the modeled propagation saw test, which results in loading of the modeled snow, has large influences on the simulation results depending on the snow properties (Gaume et al., 2015), the thicker and stronger the snow layer, the larger the column must be to get accurate results. To summarize the snow parameters important for propagation distance and speed, which are important for simulations of slab breakoff in the thesis:

- propagation speed increases if: young's modulus increases, slab depth increases, slab density increases or the slope angle increases
- propagation velocity decreases if weak layer has increasing strength
- propagation distance decreases if young's modulus, slab density or weak layer thickness increases (leads to faster failure)
- propagation distance increases if slab tensile strength, slab depth, weak layer strength or slope angle increases
- For very dense slab layers, tensile fracture and avalanche release is affected by topography (trees, large rocks in the avalanche path, or heterogeneity of snowcover)

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